China and the Environment: have we reached a turning point?

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Abstract

Globalization of the Chinese economy has contributed to the decline in environmental conditions; however there are many benefits to trade and gained prosperity that can actually improve the environment. My honors thesis uses trade data in China to consider whether a Kuznets curve relationship exists for various atmospheric pollutants. I explore their relationships using a panel of province-level trade and pollution data for various months between 2002 and 2012.

Keywords

Atmospheric pollution, Chinese trade, province-level, EKC, remote sensing

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1. Introduction

1.1 Trade Liberalization and Growth

In the 1990's China inaugurated reforms that ultimately opened their economy to the world market, dismantled the state ownership of firms, and established private property. International trade became a major focus, leading to the creation of Special Economic Zones. China embraced many aspects of the capitalist market-driven economic structure. China's state-directed industrialization led the country to become a global industrial power by 2000. China became a member of the World Trade Organization in 2001, and has free trade agreements with numerous other countries. Rapid urbanization and industrialization has increased China's need to expand its infrastructure. China is now importing large amounts of raw materials and primary products for construction. China imports many intermediate goods that will eventually be sold for export. Value-added production accounts for a large portion of trade growth.

China's exchange rate policy aided in sustained economic growth and trade expansion. The undervaluation of the Chinese Renminbi make cheap exports more attractive on the global market. A stable exchange rate also gave security to Chinese firms and their trading partners. Capital and energy-intensive industries, like iron and steel, benefited from government subsidies. These industries also benefited from Chinese reforms that led to state-owned enterprises becoming increasingly profitable. Following China's admission into the World Trade Organization, exports of apparel, textiles, and furniture increased dramatically, largely due to a natural comparative advantage in labor-intensive production. Chinese industrial policy promoted exports of high-tech goods, like cell phones, laptops, liquid crystal displays, and integrated

electronic circuits. China had already made large scale investments in these technologies as global demand soared, further contributing to an impressive Chinese domination in electronics manufacturing.

1.2 Environmental Quality

China's rapid industrialization has produced an increasing demand for energy. China has been employing cleaner-burning technology in its coal plants, and increasingly utilizes renewable energy, such as hydropower and wind. The country has seen investment in large-scale renewable energy projects, such as the Three Gorges dam and wind farms in Xinjiang. However, they are still largely dependent on fossil fuels. Coal fired plants have huge local and global environmental effects. Inferior and unwashed coal released fine particulate pollution into the air, and the inefficient burning of coal releases carbon dioxide and sulfur dioxide. China has vast reserves of coal in the north and northeast.

The environmental and health consequences of air pollution are a growing concern in the country. By 2009 China had surpassed the United States as the world's biggest emitter of carbon dioxide, increasing international scrutiny of the rapidly developed country. Environmental groups and local communities have stepped up to combat the degradation of natural resources. The government has also established an Environmental Protection Agency. China's recent Five-Year Plans include many initiatives to better the environment by increasing forest coverage, reducing pollutant discharge, reducing energy consumption per unit of GDP, and reducing population growth. There has also been a large social campaign to promote principles of conservation. The Chinese government has indicated that it is will attempt to lower

emissions, and has shown this in the commitment to invest in wind farms and other green technologies. The World Bank is also committed to supporting greener growth in China by promoting low-carbon urban transport, promoting sustainable agriculture practices, and investing in resource management systems.

2. Literature

There exists growing literature on the contribution of China's globalization to the degradation of its environment. Generally, there is a large range of environmental economic literature discussing the pollution haven hypothesis, the Environmental Kuznet's Curve, the effect of trade liberalization on pollution, and the impact of environmental regulation on trade. Results are mixed on the net effect of trade liberalization and the environment. Antweiler et al. (2001) compare levels of openness to pollution concentrations and find that greater openness is associated with small but significant decreases in pollution. Frankel and Rose (2005) control for possible endogeneity in trade policy, environmental policy and income levels. They also find openness associated with decreases in sulfur dioxide, nitrogen dioxide, and particulate matter, though the results are only significant for sulfur dioxide. Frankel and Rose find no beneficial impact of openness on carbon dioxide.

The environmental impact of production can be thought of as having three components: a composition effect, a scale effect, and a technique effect (Grossman and Kreuger 1995). The net effect of trade liberalization on the environment depends on the strength of these three terms, and if a race-to-the-bottom exists. The composition effect represents changes in emission as a result of changes in the trade structure. This depends on the comparative advantage of the country. Specialization facilitated by free trade can

promote the production of less environmentally intensive goods and promote cleaner production techniques. Trade liberalization can also have negative effects on the environment if a country specializes in highly polluting industries due to factor endowments. Cole and Elliott (2003) suggest that using differences in environmental policy to generate trade between countries find an increase in emissions after liberalization. Using differences in endowments to generate trade tends to decrease emissions after liberalization.

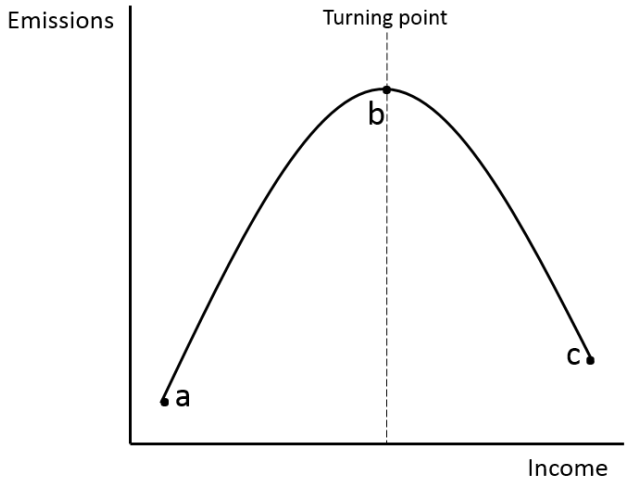
The technique effect reflects the emissions or environmental degradation associated with the production of goods. Trade liberalization can lead to improved technologies that are more energy efficient, so the production of goods and services becomes less environmentally intensive. The increase in income that trade brings can also lead people to demand better environmental quality. The technique effect is the principle emphasized by economist and policymakers as a key approach to help mitigate the impacts of climate change. The scale effect refers to the impact on the environment from the increased output resulting from freer trade. Studies find an overwhelming negative scale effect, which is the result of an increase in the demand for Chinese exports. The massive increase in production and energy use generally outweighs many of the positive effects of freer trade on the environment.

These terms are frequent in literature concerning the environmental Kuznets curve (Stern 2004, McConnel 1997), where environmental quality initially gets worse then improves with economic development. See Figure 1. From (a) to (b) we see negative scale and composition effects. The expansion of output means more resource and energy consumption resulting in more pollution. Low-income countries generally have lax

environmental regulations, therefore contributing to a comparative advantage of pollution intensive goods. However, as the wealth of a country increases there is added pressure to find cleaner production techniques and create more stringent environmental regulations. Countries will experience growth in cleaner industries. From (b) to (c) composition and technique effects begin to have positive effects on the environment. Even without trade, as an economy develops we will see a shift from agriculture to manufacturing, then later from manufacturing to services, where we would see the same curve trends.

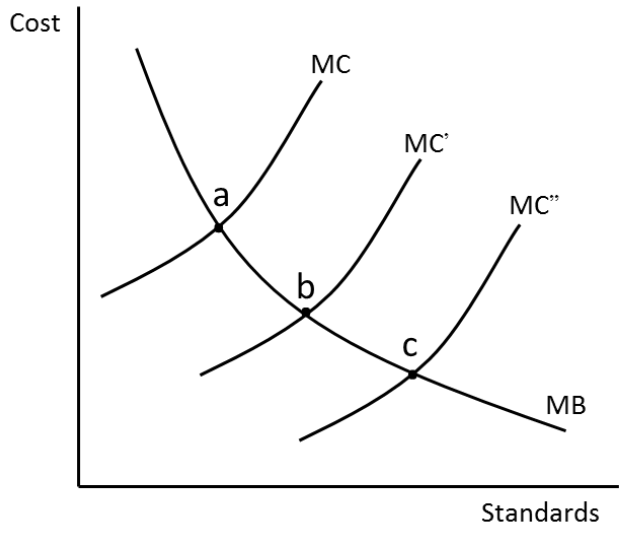
Frankel and Rose (2005) confirm the EKC hypothesis for sulfur dioxide, nitrogen dioxide, and particulate matter. Stern and Common (2001) also find an EKC controlling for time and country effects using sulfur dioxide emissions. The empirical evidence in support of a Kuznets curve relationship where pollution decreases and income increases is largely drawn from cross-country panel data on pollution and income. Variations in pollution for these applications could be attributed to different incomes and stages of development, rather than over time. There is a lack of evidence showing how pollution changes within a country.

Figure 1. Environmental Kuznets Curve



The race-to-the-bottom hypothesis frequently discussed in the literature refers to the fact that countries can engage in a competitive lowering of environmental standards to attract foreign investment by becoming pollution havens (Cole 2004, He 2006, Javorcik 2004). See Figure 2. The optimal level of environmental regulations would be where marginal benefit is equal to marginal cost (a). However, tighter regulations can drive capital flight from a country. To account for this, equilibrium moves to (b) because this loss is seen as an opportunity cost for setting high environmental standards. If other countries are lowering standards as well the opportunity cost rises once again, shifting equilibrium to (c). By lowering environmental standards, countries can create a negative composition effect where more pollution-intensive goods are being produced. Besides losing market competitiveness, countries can also be discouraged from reducing emissions due to the free-rider problem. This is suggested to be the case for carbon dioxide in Frankel and Rose (2005).

Figure 2. Race to the Bottom



Wilson et al (2002) examines 24 countries to see if environmental regulations affect exports of pollution-intensive goods. Based on a Heckscher-Ohlin-Vanek model, net exports in five pollution-intensive industries are regressed on factor endowments and measures of environmental standards. The results indicate a relationship between standards and trade. If developed countries seek to harmonize environmental standards globally at high levels, through trade agreements, then developing countries may suffer from a greater loss in exports of the pollution-intensive products than a developed country. Other empirical evidence (Levinson 2010, Dean et al. 2009) suggests that there is little influence of environmental regulations on foreign investment. For firms, pollution abatement costs often comprise a small portion of the total cost of production. Chai's study assumed that environmental standards were insignificant in the decision process for firm's location. In general, industries that are intensive in capital, energy, and land cause more environmental degradation than industries that are labor intensive. It is unlikely that China would pursue a race-to-the-bottom scenario when their comparative advantage lies in labor-intensive products. Because of this, we see a composition shift that is more beneficial for the environment.

The magnitudes of the effects explained above are an important focus of many empirical studies specific to China. Chai's study on the manufacturing industry found that trade liberalization promoted specialization in areas of comparative advantage, which included industries that contributed less to environmental degradation. In this regard, trade made a positive contribution to China's environment. Trade also enabled China to transfer environmental costs to other countries by importing intermediate products whose production contributed to environmental degradation. Overall, the scale effects offset

these gains, and China has seen tremendous environmental degradation at a rate similar to manufactured exports. Jalil and Feridun (2011) found that financial development in China has led to a decrease in environmental pollution. Authors conclude that carbon emissions are mainly determined by income, energy consumption and trade openness and their findings confirm the existence of an Environmental Kuznets Curve in the case of China.

Many studies (Chai 2002, Dean and Lovely 2010) use estimates of pollution intensities and emissions. The use of World Bank intensities, also known as the Industrial Pollution Projection System is inherently biased because it was developed using US technologies and standards. It operates through sectorial estimates of pollution intensity, or pollution per unit of activity. This database was created by merging manufacturing census data with Environment Protection Agency data on air, water, and solid waste emissions. Alternatively, there are intensities collected by China's State Environmental Protection Agency, but these are severely limited in coverage. This database does not cover firms below the county level, which ignores polluting township and village enterprises. These studies underestimate the volume of pollutants due to trade because they use intensities that do not account for a large portion of production (those coming from small producers). This will have important consequences on the composition effect, which is severely biased in both research papers.

3. Methodology

To begin exploring the possibility of a negative quadratic relationship, I model a simple EKC as:

$$S = \beta_1 \left(\frac{Y}{P}\right) + \beta_2 \left(\frac{Y}{P}\right)^2 + \varepsilon$$

(Equation 1)

Where S is a given atmospheric gas, P is population, Y is gross regional product, and ε is the random error term.

The turning point level of income, beyond which pollution begin to decline, is given by:

$$\tau = \exp(-0.5\beta_1/\beta_2).$$

(Equation 2)

Then, I estimate both fixed and random effect models with both province and year effects. The random effects assumption is that the individual specific effects are uncorrelated with the independent variables. The fixed effect assumption is that the individual specific effect is correlated with the independent variables. The EKC is expressed as:

$$s_{it} = \alpha_i + \gamma_i + \beta_1 \ln\left(\frac{Y_{it}}{P_{it}}\right) + \beta_2 \ln\left(\frac{Y_{it}}{P_{it}}\right)^2 + \varepsilon_{it}$$

(Equation 3)

Where α_i are province-specific intercepts, $\gamma_i$are time-specific intercepts, and provinces are indexed by i and specified over time t. Using a Hausman test, I will determine which model is more appropriate to use.

As discussed in the race-to-the-bottom hypothesis, countries open to trade can engage in a competitive lowering of standards. Alternatively, the gains-from-trade

hypothesis suggests that trade raises incomes and promotes technological innovation and cleaner production techniques. I will add openness to the model to consider whether trade harms or benefits the environment. Because we are looking at provinces within the country, overall trade openness is the same in terms of general policy, but some provinces may rely more heavily on trade in relation to their output. This view will be used in lieu of the concept of country openness that is found in EKC literature. Trade and the types of goods that are produced can also be influenced by province population and natural resource base. Generally labor dependent industries produce less emissions than industries that intensively use land and capital. We will account for the endowment of land per capita for all provinces. The modified EKC is expressed as:

$$s_{it} = \alpha_i + \gamma_i + \beta_1 \ln\left(\frac{Y_{it}}{P_{it}}\right) + \beta_2 \ln\left(\frac{Y_{it}}{P_{it}}\right)^2 + \beta_3\left(\frac{X_{it} + M_{it}}{Y_{it}}\right) + \beta_4 \ln\left(\frac{L_i}{P_{it}}\right) + \varepsilon_{it}$$

(Equation 4)

Where X is the sum of nominal exports, M is the sum of nominal imports, trade dependence is defined as the ratio of nominal exports and imports to gross regional product ($\frac{X_{it}+M_{it}}{Y_{it}}$), and $\left(\frac{L_i}{P_{it}}\right)$ is land area per capita.

Equations 3 and 4 follow prevalent methods of logarithmic data transformation found in EKC literature. However, because my data sources reflect observed levels of pollutants in the atmosphere and aggregated production levels, I will estimate an EKC based off Equation 4 using the variables original metric. I am also concerned that monthly seasonality exists for pollutants and trade. Atmospheric gasses like carbon dioxide naturally fluctuate with the seasons due to changes in net photosynthesis. Agricultural products also vary seasonally and can effect trade values. The model with both province and month effects is expressed as:

$$s_{it} = \alpha_i + \gamma_i + \beta_1 \left(\frac{Y_{it}}{P_{it}}\right) + \beta_2 \left(\frac{Y_{it}}{P_{it}}\right)^2 + \beta_3 (\frac{X_{it} + M_{it}}{Y_{it}}) + \beta_4 \left(\frac{L_i}{P_{it}}\right) + \varepsilon_{it}$$

(Equation 5)

When Equations are estimated with province and time dummies, the dummies for one country and one time interval should be dropped. The results will be estimated relative to the omitted province and time.

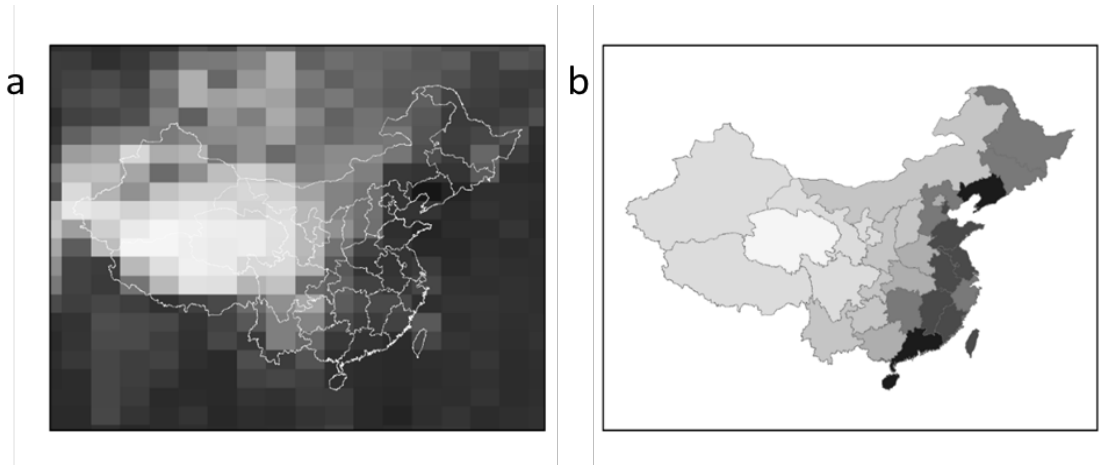
4. Data

Even though ground and airborne measurements of pollutants have existed since 1960, these data contain large spatial gaps. This issue can only be solved by using atmospheric observations from space. The complexity of such a task, however, has made them possible only very recently. Environmental variables for the time series were generated using remotely sensed imagery. NASA's satellites Aura and Aqua are home to many sensors that provide long-term monitoring of atmospheric chemistry. See Table 1 for variables obtained. These sensors focus on the troposphere, the layer of atmosphere that stretches from the ground to the altitude at which airplanes fly. Provinces in China have been given mean values of all the pixels that fall within that provinces boundary. This raster to vector process is visualized in Figure 3.

Table 1: Spatial Data

Variables	Sensor	Time Period	Units
Carbon Dioxide Mole Fraction	Atmospheric Infrared Sounder	September 2002 – February 2012	Parts per Million
Total Column Methane	Tropospheric Emission Spectrometer	September 2004 – June 2011	Mole per cm2
Total Column Carbon Monoxide	Tropospheric Emission Spectrometer	September 2004 – June 2011	Mole per cm2
Ozone Total Column Density	Tropospheric Emission Spectrometer	September 2004 - December 2009	Dobson Unit
Ozone Tropospheric Column	Tropospheric Emission Spectrometer	September 2004 - December 2009	Dobson Unit

Figure 3. Raster to Vector transformation.



Map (a) presents the original format with China's provinces overlaid on top. In map (b), every province is given the average value of pixels found within the boundary.

Carbon dioxide fraction indicates the amount of CO2 in the atmosphere by volume. When expressed in PPM, it is the number of molecules of CO2 represented in a million atmospheric gas molecules. Total Column Methane is the atmospheric density of methane in a vertical column of air. Total column concentration of carbon monoxide is

the number of molecules of CO in an atmospheric column from the Earth's surface to the top of the stratosphere above a square centimeter of the surface. Total Column Ozone is the atmospheric density of ozone in a vertical column of air. The Dobson Unit is a measure of total ozone. See Appendix A for more information regarding this remotely sensed data.

Carbon dioxide is the most prominent greenhouse gas. A byproduct of fossil fuel combustion and biomass burning, carbon dioxide is also emitted from land use changes and other industrial processes. Figure 4 shows the distribution of Carbon Dioxide by province. Carbon dioxide fraction varies from region to region as it is constantly being released into the atmosphere, taken out of the atmosphere, being mixed with other atmospheric gases, and dispersed by winds. While globally the amount of carbon dioxide in the atmosphere is increasing, the amount of carbon dioxide present in the atmosphere over any particular region during any particular time period will both decrease and increase variably.

Methane is a very powerful greenhouse gas, but easily breaks down in the atmosphere. Human activities that contribute to methane emissions include natural gas and petroleum systems, landfills, coal mining, raising livestock and manure management. Concentrations of methane in the atmosphere may be the result of large quantities of fossil fuel combustion or biomass burning at the Earth's surface and will likely be concentrated in these areas. Figure 5 shows methane observed over time in China.

Like methane, Carbon monoxide is a byproduct of combustion. Carbon monoxide absorbs infrared radiation just like carbon dioxide and methane, but because carbon monoxide molecules have a short atmospheric lifetime, they have little influence on

global warming. Carbon monoxide may be a weak direct greenhouse gas, however carbon monoxide indirectly increases the global warming potential of other gases by reducing the abundance of hydroxyl radicals in the atmosphere that reduce the lifetimes of many strong greenhouse gases. Carbon monoxide is also an excellent tracer for pollution sources and pollution pathways through the troposphere. The global distribution of carbon monoxide closely resembles its source distribution, which is greatest near the surface. Figure 6 shows carbon monoxide distribution across China.

Tropospheric ozone is another greenhouse gas. Tropospheric ozone is a secondary pollutant which is not emitted directly into the atmosphere. Volatile organic compounds, nitrogen oxides and sulfur dioxide in the troposphere create smog, which then reacts with sunlight to produce ozone. Therefore, tropospheric ozone will be more noticeable over urban areas and on warm days. Tropospheric ozone also can decrease the amounts of methane and other greenhouse gases because it reacts with water vapor to form hydroxyl radicals. This type of ozone pollution can be visualized over time in Figure 7.

Ozone in the stratosphere absorbs harmful ultraviolet light. As the troposphere warms from more intensive greenhouse effects, the stratosphere cools. This is because a greater proportion of the outgoing heat energy that would usually go into the stratosphere gets trapped instead in the troposphere due to the greenhouse gases. Colder temperatures in the stratosphere can result in a thinning of the ozone layer. CFC emissions, which have been banned in the Montreal Protocol, also depleted stratospheric ozone, increasing the risk of skin cancer. Figure 8 shows the trend of ozone density over time.

Economic data includes firm level import and export, Gross Regional Product and population. Figures 9 and 10 show trade values aggregated by year for each province. Table 2 presents a detailed summary of economic variables and their sources.

Table 2. Economic Data

Variables	Source	Time Period	Units
Gross Regional Product	National Bureau of Statistics of China	December 2001-December 2013	100 million Yuan
Resident Population	National Bureau of Statistics of China	December 2001-December 2013	10 thousand persons
Exports	Chinese customs office	January 2002-December 2006	Nominal US dollars
Imports	Chinese customs office	January 2002-December 2006	Nominal US dollars

Population and Gross Regional Product were obtained from the National Bureau of Statistics of China. The year-end resident population refers to the population number at 24 pm. 31st December. Population data for 2000 and 2001 are estimated on the basis of population census, the rest of the data are estimated on the basis of the annual national sample surveys on population changes. Gross regional product data applies to "Industrial classification for national economic "(GB/T4754-2002) in 2004 and after. Gross regional product data applies to "Industrial classification for national economic "(GB/T4754-1994) before 2004. Monthly values were generated for these variables using linear interpolation. Firm level import and export, disaggregated by firm type and trade regime was sourced from the Chinese customs office.

5. Results

5.1 Pollution Trends

Table 3 displays correlation coefficients for atmospheric pollutants and economic variables. There are weak relationships between many of the variables, however the signs are very telling. Intuitively, one would image a positive relationship between land per capita and the concentration of pollution. If land per capita is low then there is a greater abundance of people with limited space. This would encourage labor-intensive manufacturing and promote a service-based economy. If land per capita is high then production will be influenced by the abundance of land and produce land and resource-intensive goods. I would imagine the latter type of production is more polluting, but the scale effect of urbanization in china may affect this. There is a moderate negative relationship between methane and land endowment.

One would also expect pollutants to increase with gross regional product due to the scale effect of production. There is a moderate positive linear relationship between methane and gross regional product. The signs for exports and imports are interesting and can reveal information about whether China's comparative advantage is in clean or dirty goods. If exports are negatively related then China is specializing in cleaner goods. If exports are positive it suggests that China is specializing in dirty goods. The positive sign of trade dependence on the majority of pollutants suggests that trade liberalization results in more atmospheric pollutant for China, however the magnitudes are not convincing.

Table 3. Pearson's Correlation Coefficients

	Carbon Monoxide	Carbon Dioxide	Methane	Trop. Ozone	Ozone Density
Export	0.2536	-0.0287	0.3546	0.1175	-0.1204
Import	0.2671	-0.0123	0.3648	0.1469	-0.0892
Land Endowment	-0.4246	-0.0231	-0.4936	-0.2705	-0.0792
Trade Dependence	0.3161	-0.0364	0.4206	0.2019	0.0022
Gross Regional Product	0.4141	0.1207	0.5082	0.336	-0.0491

Values for the following maps across time are displayed as averages for an entire year to account for natural fluctuations of atmospheric gasses due to weather and climate. In Figure 4 we see a sharp rise of carbon dioxide over the past 10 years, with levels being higher in northern and coastal provinces. It is also important to note that the ppm values themselves are incredibly high and unprecedented. Average methane is shown in Figure 5. For methane we see higher values along the eastern coast. Figure 6 shows Average carbon monoxide. This pollutant is most prominently located in coastal and southeast provinces. Figure 8 shows a decline of Ozone density over time, where deterioration began in southern provinces. Northeast provinces, like Heilongjiang, Jilin, and Inner Mongolia have sustained levels of average column density. Tropospheric Ozone is rising in central coastal provinces. This increase tells us that there is greater human induced ozone pollution in these regions. This pollutant is concentrated in urbanized areas.

Figure 9 shows the aggregated value of exports for each province in a given year. The highest exporting provinces are those along the southern coastline, from Guangdong to Tianjin. The central provinces of Tibet and Qinghai have relatively low values of trade over time. The import trend closely follows the distribution of exports, this is likely

because China imports many intermediate goods that will eventually be resold for export. Trade is also concentrated in the most heavily populated and urbanized provinces, like Jiangsu, Guangdong, Zhejiang, and Shandong.

Figure 4. Carbon Dioxide

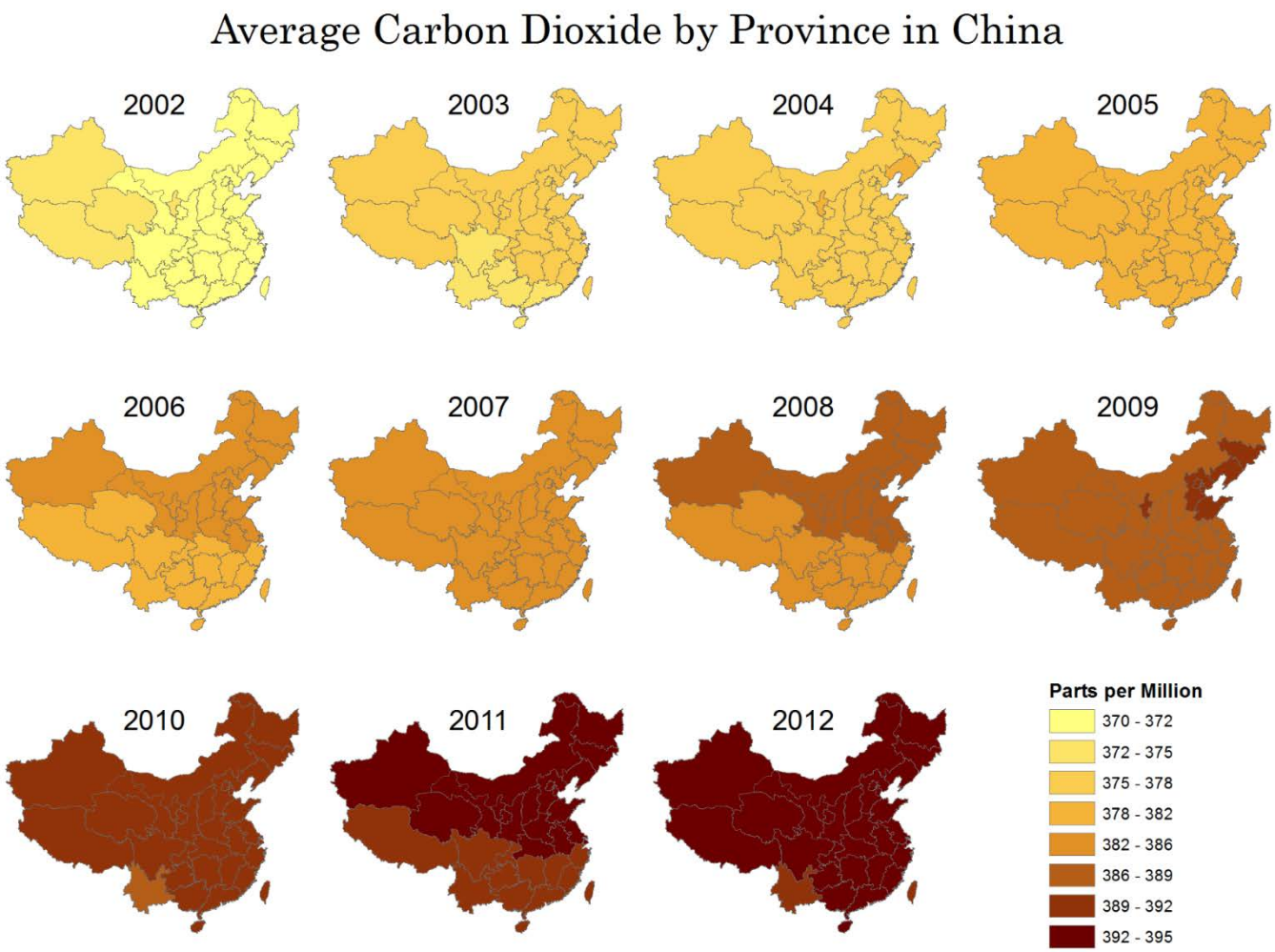


Figure 5. Carbon Monoxide

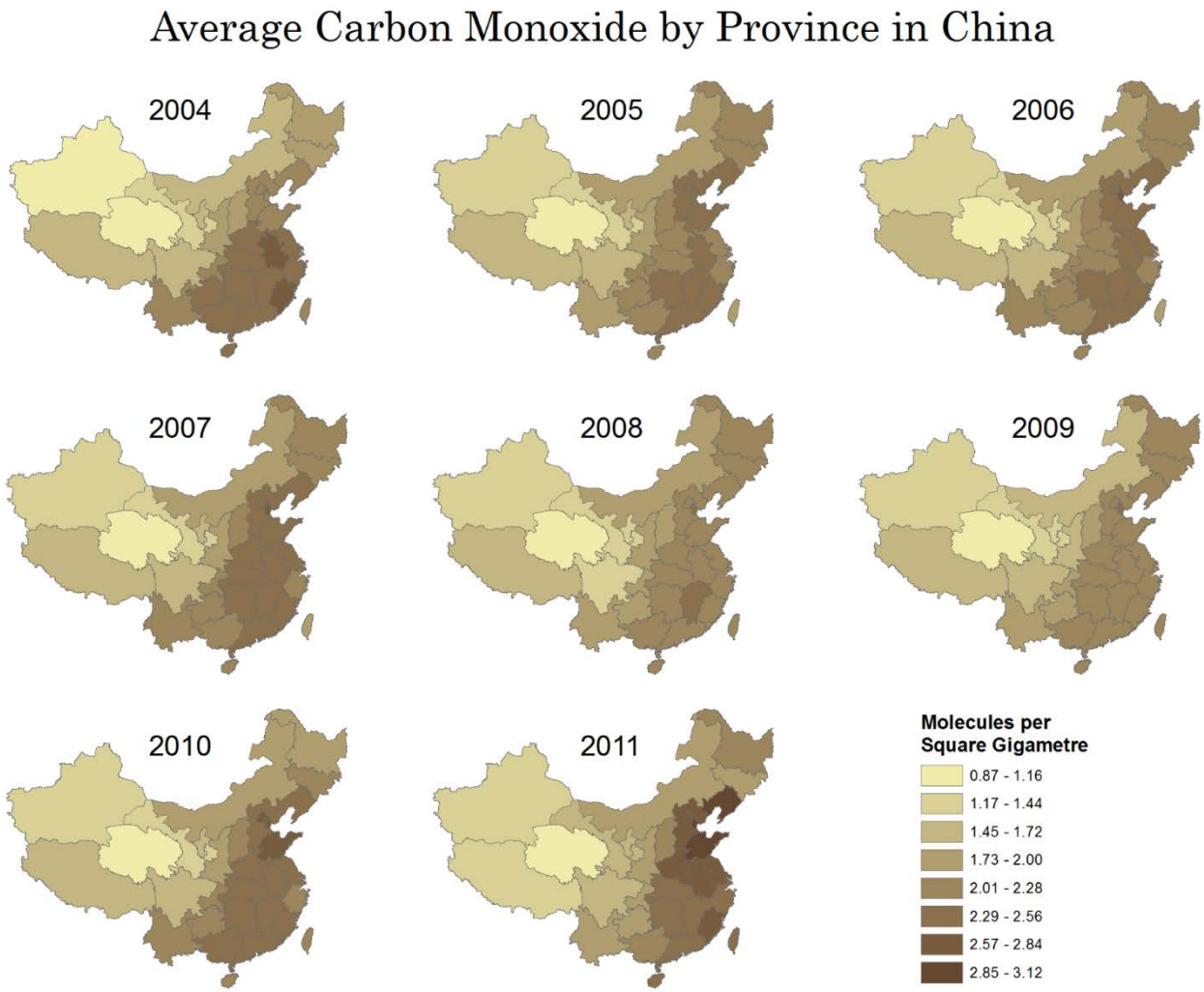


Figure 6. Methane

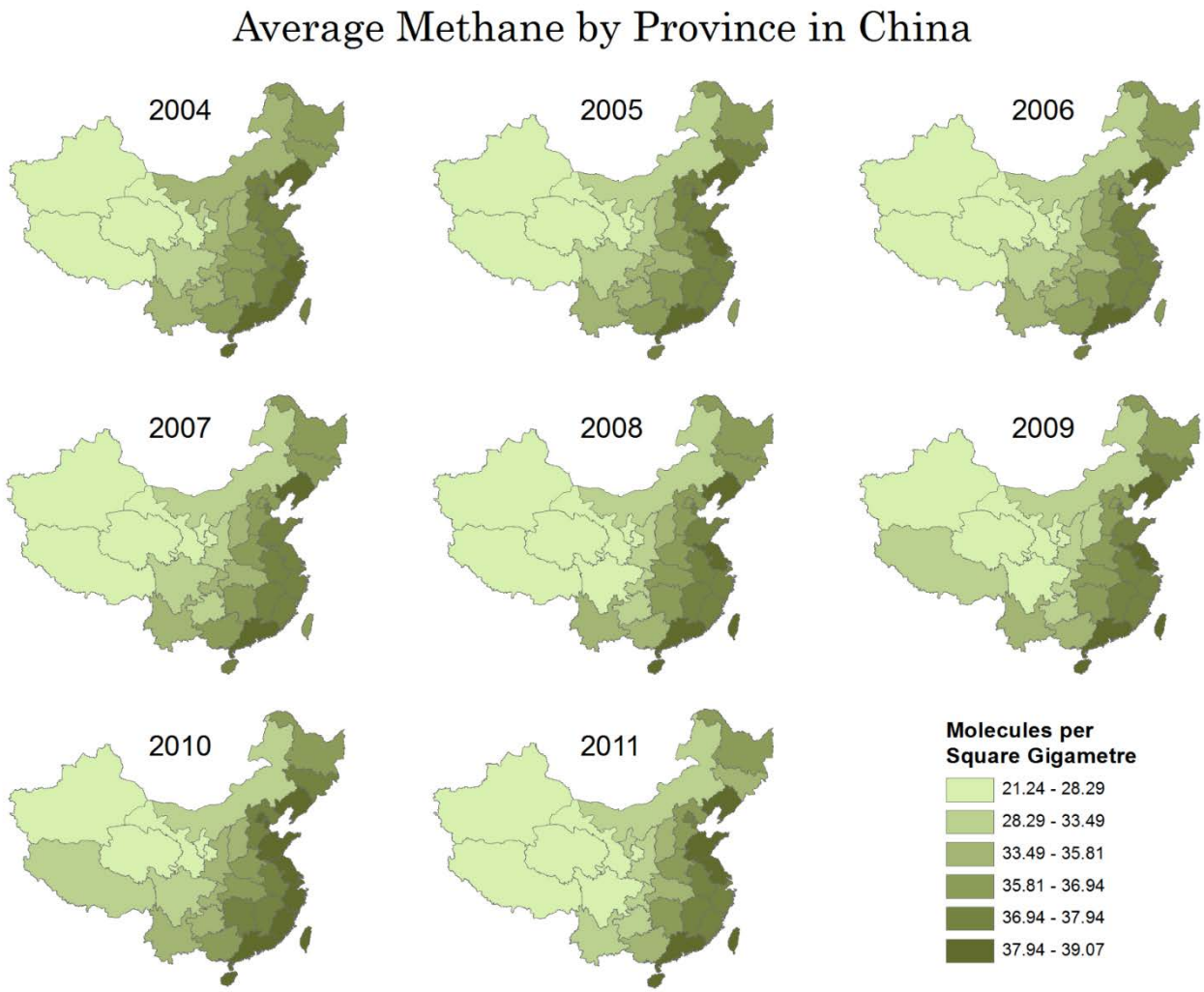


Figure 7. Tropospheric Ozone

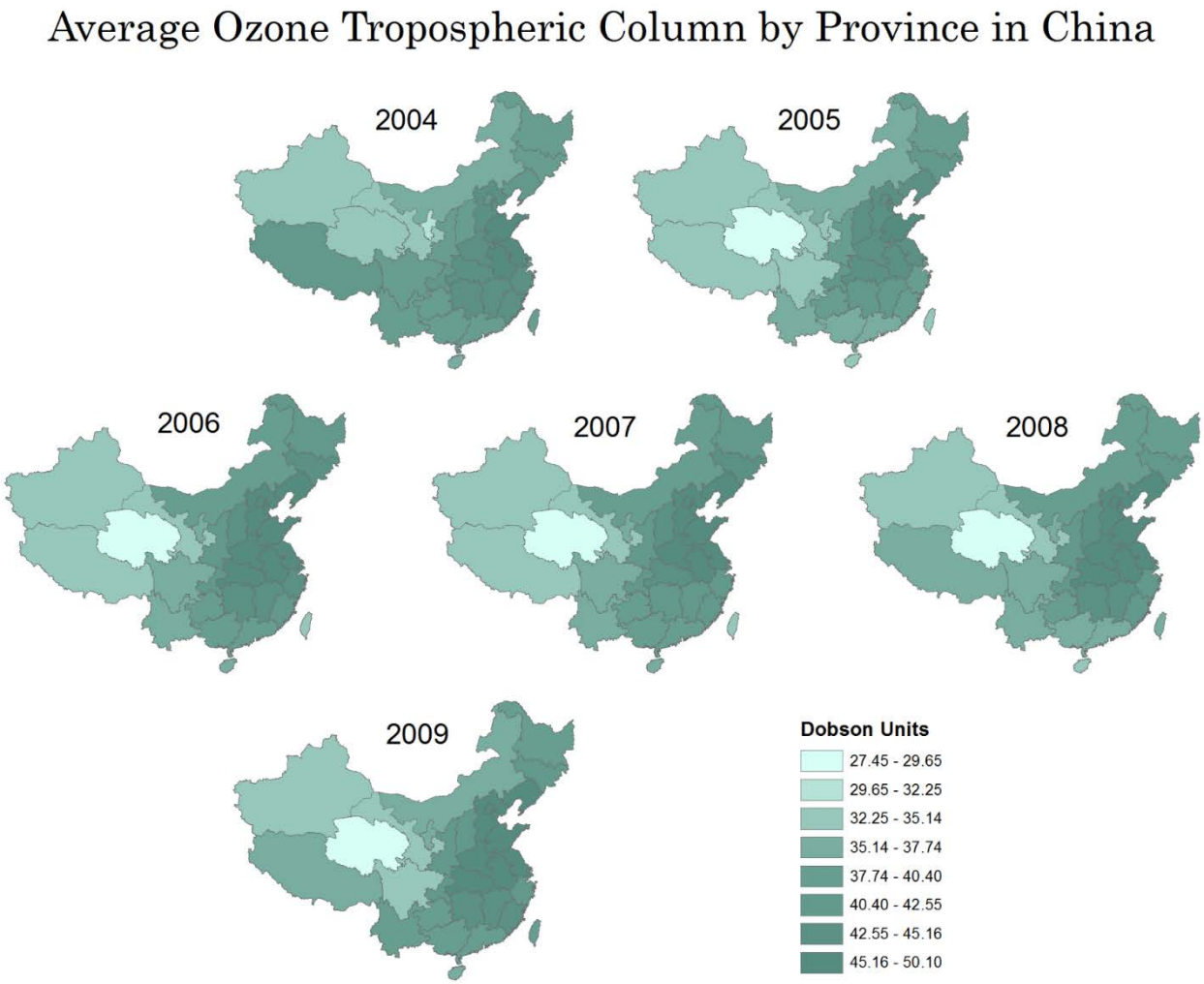


Figure 8. Ozone Density

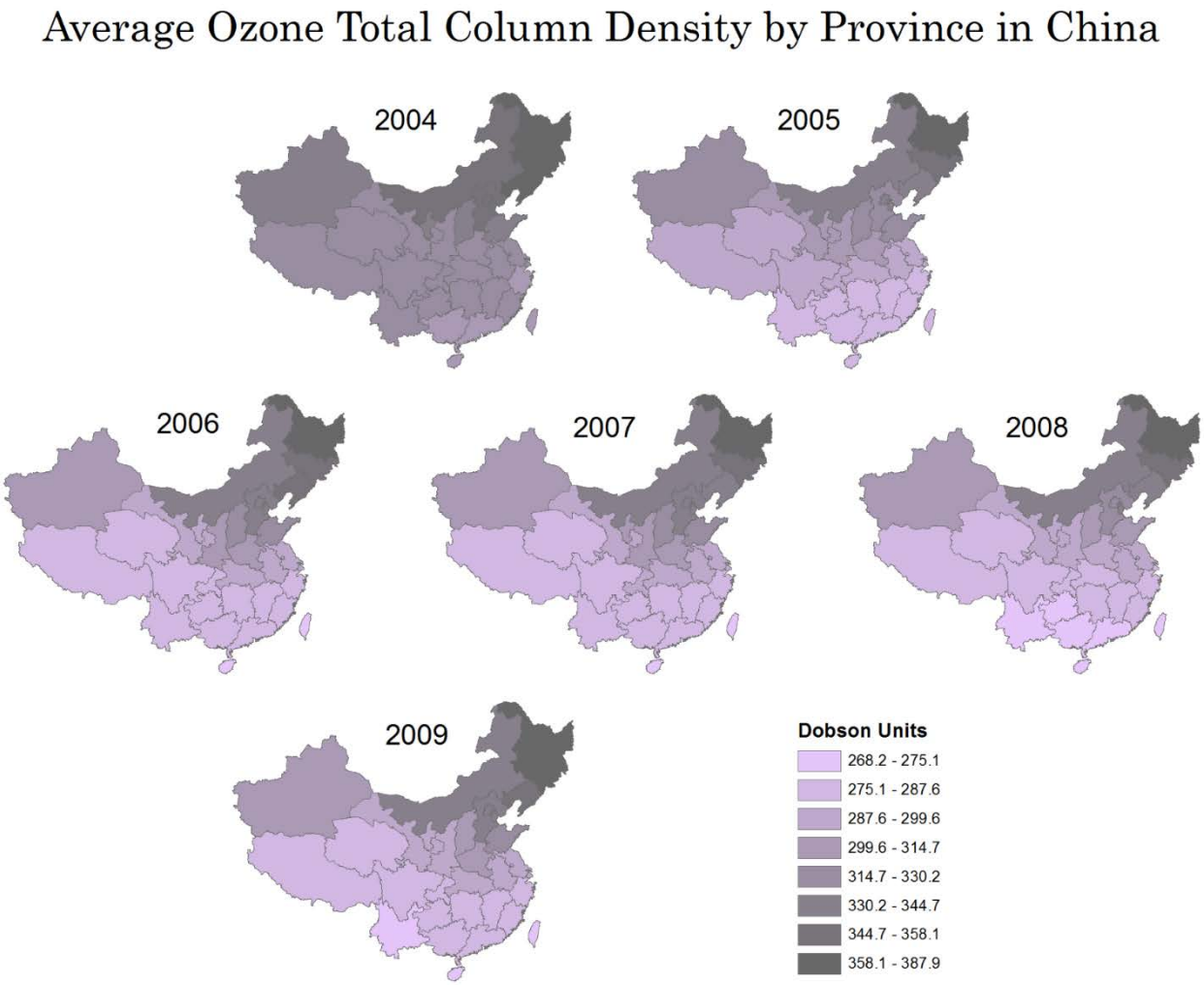


Figure 9. Exports

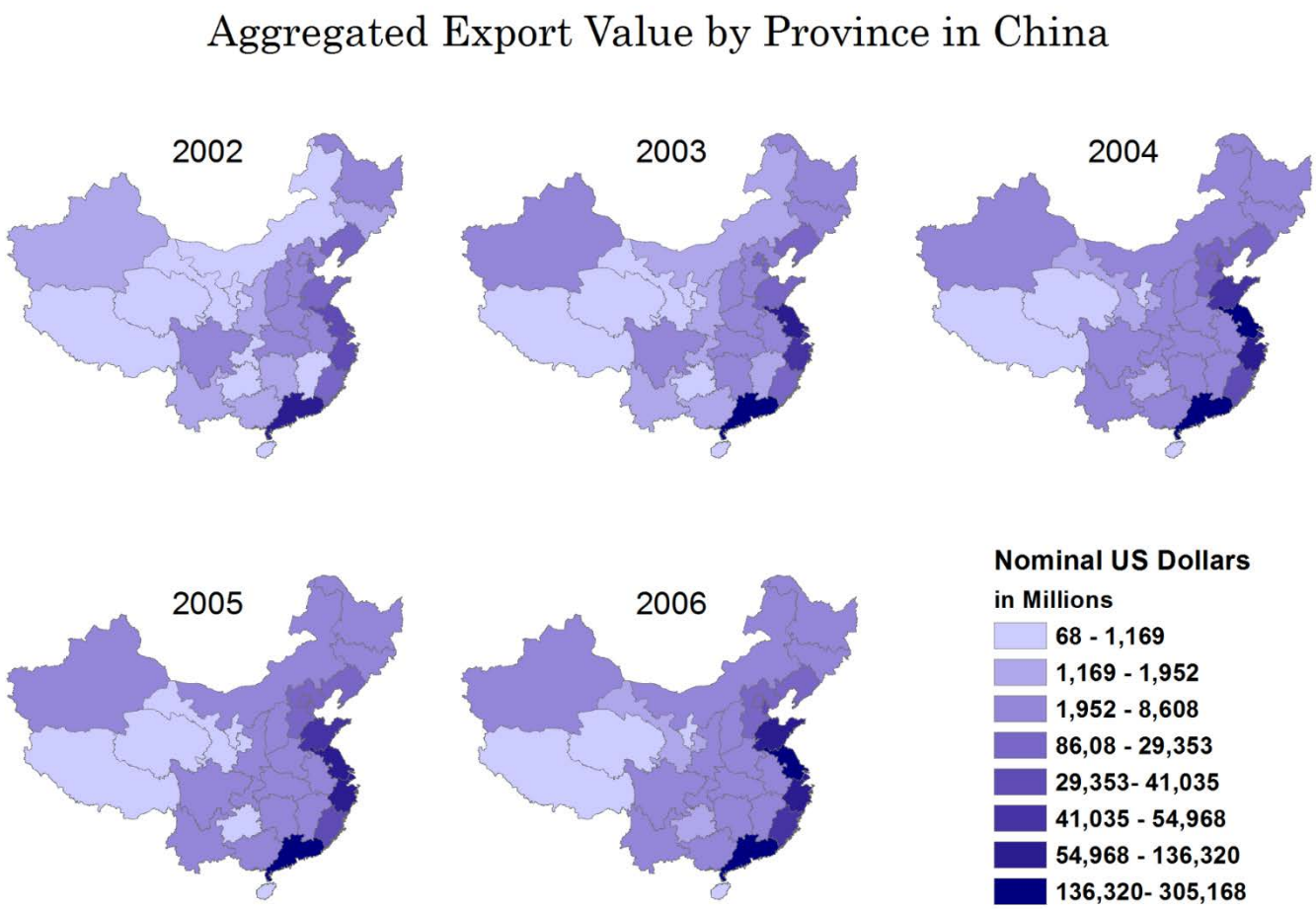
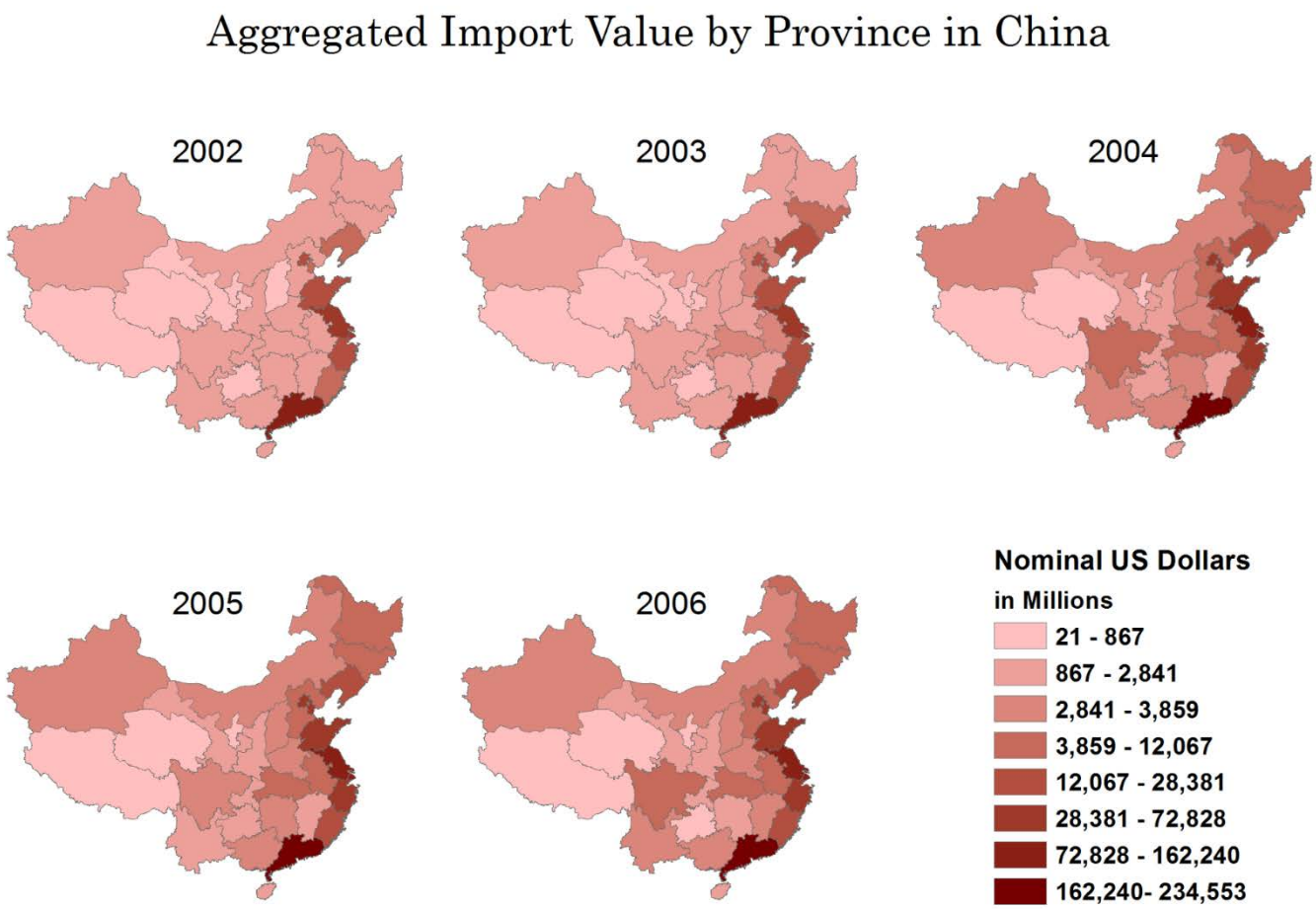


Figure 10. Imports



5.2 EKC Modeling

The regression results for Equation 1 are presented as graphs in Figure 11. This allows us to see if the hypothesized Kuznets curve accurately reflects the relationship of these pollutants to product per capita. It is important to stress that we are looking for an inverted U-shape where product per capita has a positive sign, and squared product per capita has a negative sign, with the exception of ozone density which should have a U-shape. This is because density can be thought of as an environmental "good" where high density means lesser amounts of harmful UV rays. All coefficients are significant at the .01 level. All parameters, with the exception of ozone density, have the expected sign, where pollution increases with product per capita and decreases with squared product per capita. It appears that a turning point is in fact taking place at higher levels of GRP per capita. Using Equation 2, τ is 52,824 yuan per capita for carbon dioxide; 54,007 for methane; 72,000 for carbon monoxide; 55,655 for tropospheric ozone; and 42,176 for ozone density.

We now look to expand this model and account for fixed and random effects as well as other variables that may influence our atmospheric pollutants. The results for Equation 3 are presented in Table 4. The year effect is estimated relative to 2002 for carbon dioxide and 2004 for carbon monoxide, methane, tropospheric ozone columns, and ozone total column density. The province effects in the random model are estimated relative to Beijing. The parameters for production per capita and squared production per capita have unexpected signs for all pollutants, with the exception of methane. The former pollutants have negative income effects with positive squared income. However, few of these coefficients are statistically significant.

If there are omitted variables, and these variables are correlated with the variables in the model, then fixed effects models provides a control for omitted variable bias. Provinces will serve as their own controls. This assumes that the effects of an omitted variable would be constant over time for every province. A Hausman test for bias or inconsistency of an estimator will determine which model is appropriate to use. Under the null, estimators will have no systematic difference. The initial hypothesis that province-level effects are adequately modeled by a random-effects model is rejected for Carbon Dioxide, Carbon Monoxide, Tropospheric Ozone, and Ozone Density. We cannot reject the null hypothesis for Methane. In this case, there are no significant differences between the parameters of the fixed and random effects models.

Table 5 shows the fixed and random effects models using Equation 4, where we have expanded the model to include variables for trade dependence and land endowment. The Hausman test statistic suggests that the fixed effects and random effects model have inconsistent estimators. Therefore, we reject the null and assert that the fixed effect model is a better estimator. There are no systematic differences between the fixed and random effects model for Methane. There is likely omitted variable bias, even in this expanded EKC model, so we will continue with the fixed effect approach. To confirm an EKC relationship we must observe a negative quadratic term for product per capita. However, we have no significant results in Table 5 that fit the theory. We also fail to find results that are statistically significant from zero for our trade dependence variable. To account for unobserved characteristics that may be occurring, the time-specific interval will be changed from year to month.

Table 6 presents the results for Equation 4 using month effects. This model produces significant results at the .05 level, confirming an inverted U-shaped relationship with a positive coefficient for product per capita and negative coefficient for squared product per capita. We are interested in the coefficient for trade dependence. For carbon dioxide, trade dependence is negative and significant at the .01 level. This indicates that any adverse race-to-the-bottom effect would be outweighed by positive gains-from-trade effect. Land per capita has a negative effect on carbon dioxide and is also significant at the .01 level. Therefore, carbon dioxide decreases as land per capita increases. For ozone density we also confirm our stated theory of a U-shape relationship with a negative coefficient for product per capita and positive square term, that are both significant at the .01 level. Trade dependence and land per capita also have reversed signs for ozone density. Density increases with trade dependence, indicating positive gains-from-trade. Ozone density also increases with land per capita. The results for other atmospheric pollutants are not so convincing of the EKC hypothesis, and most estimates are not significantly statistically different from zero.

Table 7 presents the results for Equation 5 where variables are expressed in their original metric. The turning point for carbon dioxide is 52,676 yuan per person. The turning point for ozone density is 49,206. These values are slightly higher than estimated in Equation 1. The regression results also suggest that there is no discernable Kuznets curve relationship for carbon monoxide, methane, or tropospheric ozone.

Figure 11. EKC results using Equation 1.

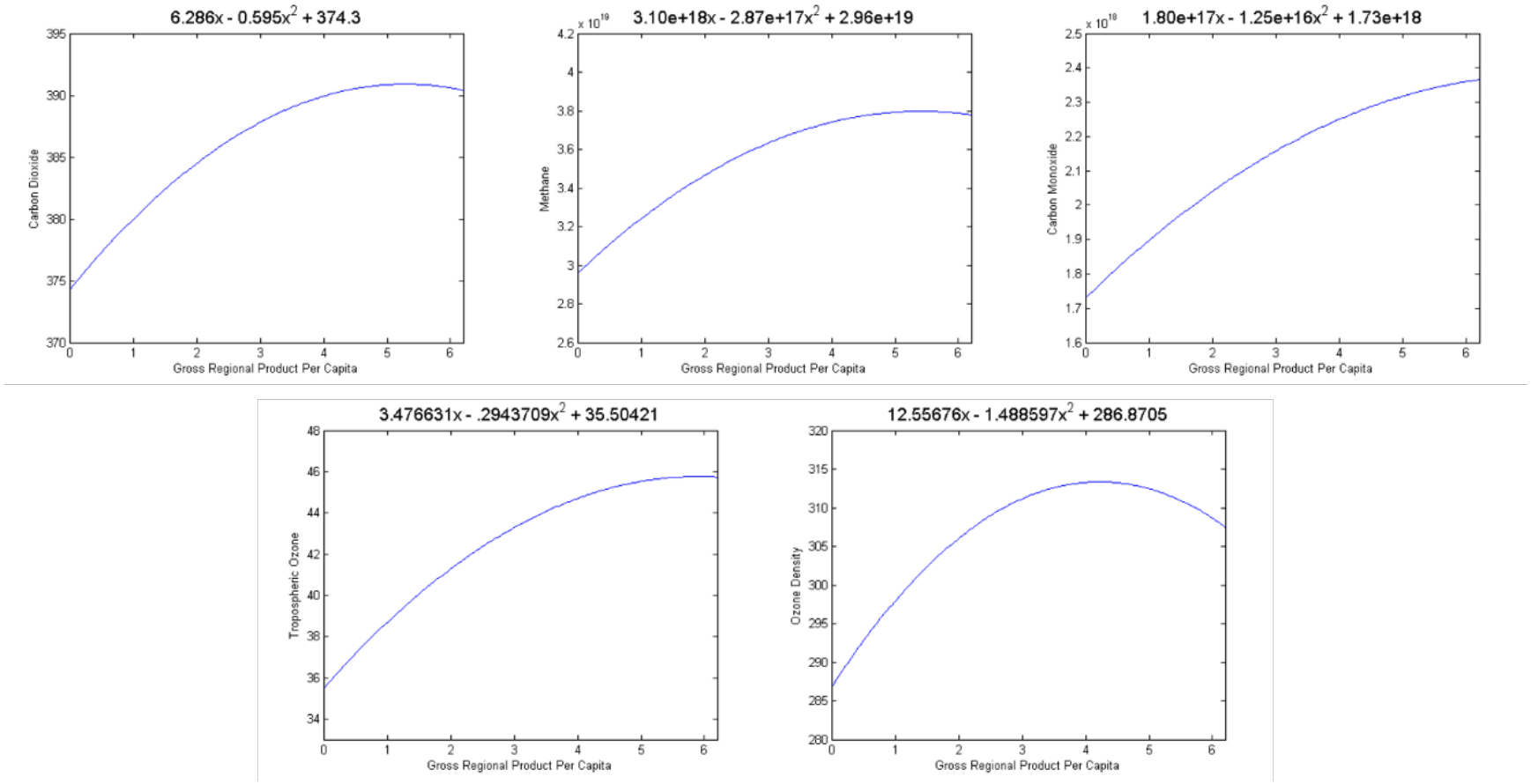


Table 4. Equation 3 Results using year time effects.

Pollutant	Carbon Dioxide		Carbon Monoxide		Methane		Tropospheric Ozone Columns		Ozone Total Column Density	
Model	Fixed Effects	Random Effects	Fixed Effects	Random Effects	Fixed Effects	Random Effects	Fixed Effects	Random Effects	Fixed Effects	Random Effects
Constant	371.2	371.3	2.216e+18	2.161e+18	3.467e+19	3.464e+19	42.02	40.38	349.6	338.7
	(0.123)	(0.171)	(2.453e+16)	(6.950e+16)	(7.884e+16)	(7.104e+17)	(0.590)	(0.955)	(2.346)	(5.047)
$\ln\left(\frac{Y_{it}}{P_{it}}\right)$	-1.061	-0.217	-8.565e+17	-5.559e+17	4.523e+17	5.462e+17	-11.87	-2.790	-118.8	-58.35
	(0.258)	(0.183)	(7.670e+16)	(6.782e+16)	(2.457e+17)	(2.425e+17)	(2.009)	(1.370)	(7.992)	(6.371)
$\ln\left(\frac{Y_{it}}{P_{it}}\right)^2$	-0.00394	0.0634	-2.108e+16	1.560e+16	2.725e+17	2.829e+17	0.203	1.192	-5.431	1.125
	(0.0600)	(0.0573)	(2.200e+16)	(2.175e+16)	(7.048e+16)	(7.035e+16)	(0.598)	(0.565)	(2.378)	(2.357)
$Adj\ R^2$	0.957		0.134		0.041		0.016		0.185	
N	3,534	3,534	2,076	2,076	2,075	2,075	1,860	1,860	1,860	1,860
AR(1)χ^2(1)	.8676888	.8352224	.9505401	.9255518	.9849225	.9847459	.9343051	.8687773	.9695036	.9240627
Hausman test	24.12 (0.0196)		68.56 (0.0000)		5.16 (0.8197)		38.66 (0.0000)		146.52 (0.0000)	

Figures in parentheses are standard errors for fixed effect regression coefficients and z-statistics for random effect regression coefficients. Figures in parenthesis for the Hausman test statistic are significance levels.

Table 5. Equation 4 results using year time effects.

Pollutant	Carbon Dioxide		Carbon Monoxide		Methane		Tropospheric Ozone Columns		Ozone Total Column Density	
Model	Fixed Effects	Random Effects	Fixed Effects	Random Effects	Fixed Effects	Random Effects	Fixed Effects	Random Effects	Fixed Effects	Random Effects
Constant	214.2 (58.25)	371.1 (1.578)	-3.108e+19 (1.911e+19)	6.259e+18 (6.521e+17)	8.276e+19 (7.090e+19)	7.307e+19 (6.068e+18)	-1,215 (434.3)	85.74 (7.808)	-9,407 (1,833)	394.1 (62.25)
$\ln\left(\frac{Y_{it}}{P_{it}}\right)$	-2.721 (0.641)	0.230 (0.278)	-8.611e+17 (1.896e+17)	-2.550e+17 (1.161e+17)	1.781e+18 (7.012e+17)	1.710e+18 (5.876e+17)	-25.33 (4.309)	-0.454 (1.790)	-285.9 (18.18)	-51.43 (11.82)
$\ln\left(\frac{Y_{it}}{P_{it}}\right)^2$	0.528 (0.227)	0.162 (0.155)	3.436e+17 (1.099e+17)	1.550e+17 (6.696e+16)	-3.206e+17 (4.076e+17)	-3.974e+17 (2.851e+17)	8.927 (2.498)	1.289 (1.182)	45.44 (10.54)	15.36 (7.014)
$\ln(\frac{X_{it}+M_{it}}{Y_{it}})$	2.17e-07 (3.14e-07)	-1.20e-07 (2.51e-07)	-3.724e+10 (9.148e+10)	-6.915e+10 (7.737e+10)	-3.154e+11 (3.383e+11)	-3.627e+11 (3.256e+11)	1.91e-06 (2.08e-06)	-1.87e-06 (1.36e-06)	2.01e-05 (8.77e-06)	8.05e-06 (8.10e-06)
$\ln\left(\frac{L_i}{P_{it}}\right)$	8.787 (3.266)	0.0162 (0.0876)	1.866e+18 (1.072e+18)	-2.330e+17 (3.570e+16)	-2.693e+18 (3.979e+18)	-2.146e+18 (3.367e+17)	70.52 (24.37)	-2.511 (0.423)	548.0 (102.8)	-3.662 (3.401)
$Adj\ R^2$	0.834		0.008		0.008		0.040		0.329	
N	1,612	1,612	775	775	774	774	775	775	775	775
AR(1)χ^2(1)	.9966791	.8661146	.9948778	.7262253	.9051097	.8962879	1.002389	.629477	.998366	.7679196
Hausman test	33.27 (0.0000)		20.07 (0.0012)		3.67 (0.5980)		42.53 (0.0000)		311.57 (0.0000)	

Figures in parentheses are standard errors for fixed effect regression coefficients and z-statistics for random effect regression coefficients. Figures in parenthesis for Hausman test statistic are significance levels.

Table 6. Results for Equation 4 using month time effect.

Pollutant	Carbon Dioxide	Carbon Monoxide	Methane	Tropospheric Ozone Columns	Ozone Total Column Density
Constant	751.8 (38.84)	-2.843e+18 (1.708e+19)	1.462e+19 (6.235e+19)	-481.6 (344.3)	-4,080 (1,642)
$ln\left(\frac{Y_{it}}{P_{it}}\right)$	13.27 (0.133)	-5.183e+16 (8.369e+16)	-4.127e+17 (3.052e+17)	-5.139 (1.688)	-124.0 (8.048)
$ln\left(\frac{Y_{it}}{P_{it}}\right)^2$	-0.379 (0.155)	2.941e+17 (1.003e+17)	-2.257e+17 (3.662e+17)	7.691 (2.023)	39.95 (9.649)
$(\frac{X_{it}+M_{it}}{Y_{it}})$	-6.60e-07 (1.94e-07)	4.795e+10 (8.368e+10)	-4.335e+11 (3.044e+11)	3.15e-06 (1.69e-06)	4.24e-05 (8.05e-06)
$ln\left(\frac{L_i}{P_{it}}\right)$	-21.10 (2.179)	2.701e+17 (9.582e+17)	1.151e+18 (3.499e+18)	29.12 (19.32)	247.1 (92.15)
R-squared	0.917	0.183	0.209	0.377	0.444

Figures in parentheses are standard errors

Table 7. Results for Equation 5 using month time effect.

Pollutant	Carbon Dioxide	Carbon Monoxide	Methane	Tropospheric Ozone Columns	Ozone Total Column Density
Constant	364.4 (0.608)	2.033e+18 (2.702e+17)	3.582e+19 (9.790e+17)	23.22 (5.401)	355.4 (26.70)
$\left(\frac{Y_{it}}{P_{it}}\right)$	14.96 (0.226)	1.147e+16 (9.603e+16)	-2.836e+17 (3.479e+17)	-3.263 (1.920)	-105.4 (9.489)
$\left(\frac{Y_{it}}{P_{it}}\right)^2$	-1.420 (0.0396)	1.987e+16 (1.550e+16)	-2.108e+16 (5.614e+16)	0.819 (0.310)	10.71 (1.532)
$(\frac{X_{it}+M_{it}}{Y_{it}})$	-8.91e-07 (2.44e-07)	2.648e+10 (8.375e+10)	-4.652e+11 (3.032e+11)	2.84e-06 (1.67e-06)	3.58e-05 (8.28e-06)
$\left(\frac{L_i}{P_{it}}\right)$	-1.11e-08 (1.86e-09)	-2.023e+08 (8.162e+08)	-1.385e+09 (2.954e+09)	6.23e-08 (1.63e-08)	2.59e-07 (8.06e-08)
R-squared	0.867	0.176	0.210	0.383	0.408

Figures in parentheses are standard errors

5.2 Problems with the EKC

The EKC may fail to describe emission trends for all pollutants due to differences in pollutant sources, half-life, human awareness or exposure, etc. The EKC model can also fail to fully explain pollution because the atmosphere is an open access resource. It is also difficult to allocate pollution to one province, or even country. Atmospheric pollutants transcend the spatial extent of arbitrary political boundaries. While actual pollution concentrations were taken over each province, they may be sourced from elsewhere. The satellite that remotely senses these atmospheric products is set up to observe in the mid- to upper troposphere, so values detected in a certain column of the atmosphere may have been brought there by weather, instead of from events on the ground.

Trade also allows heavily polluting production and resource intensive goods to move elsewhere, which doesn't lower global observations of pollution. Countries can reduce their domestic emissions by shifting to the production of less resource-intensive goods and services while increasing the pollution intensity of their imports. This does not necessarily result in any worldwide decline in pollution, and perhaps even increases global pollution if the producers of the imports have less efficient production technologies.

It is standard for EKC cross-country studies to account for polity. Frankel and Rose (2005) find that improved democratic governance has a beneficial effect on the environment. The EKC theory where at high levels of income people begin to demand higher environmental standards is dependent on the political influence of citizens. The political structure of China can inhibit the possibility of an EKC.

The EKC is an inadequate model and estimates of EKC income levels can suffer from significant omitted variable bias. The Hausman test shows significant difference in parameter estimates for the random-effects and fixed-effects model. This suggests that our regression variables are correlated with the country and time effects. There are likely omitted variables that vary across provinces or across time leading to bias in the model. Future research should include tests for the possibility of a unit root and cointegration.

6. Conclusion

Advances in climate change science and remote sensing have allowed us to observe atmospheric gasses with high accuracy and extensive spatial coverage. The primary contribution of this paper is to address concerns in data quality that results in misinterpreted empirical studies. Within China there is evidence to suggest that an environmental Kuznets curve may exist for carbon dioxide and ozone density. However, I have no evidence to support this claim for the other pollutants tested.

Trade appears to have beneficial effects, however these conclusions are conditional on the model used in the analysis. The fixed effect model explains more of the variation in pollutants and has better properties than the random-effects model. There are likely many omitted variables that can contribute to the robustness of models. I also reject that a race-to-the-bottom exists in China.

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Appendix A

AIRS gridded products are released in a 1°x1° spatial resolution. Monthly products will likely contain complete global coverage without gores (cells with no data). This instrument orbits 15 times a day to create a global survey. The monthly L3 products address the interests of those involved in climate trend analysis and low-frequency climate variability. They are typically interested in monthly means over long timescales and prefer data products with the lowest possible systematic errors. NASA/NOAA estimates the uncertainty of these measures at 1-2 ppm.

The Goddard Earth Sciences Data and Information Services Center at NASA Goddard Space Flight Center is archiving and distributing all AIRS data. All of these data records are searchable by various criteria (temporal, spatial, event), via the Mirador search engine (http://mirador.gsfc.nasa.gov/), and are available online for free download by FTP.

TES Monthly L3 products are released in a 4°x2° spatial resolution. Monthly products represent Global Surveys that are initiated within that calendar month. The routine TES observation mode is to produce global survey standard products spanning 16 orbits on a 50% duty cycle, or approximately every other day. The primary objective of TES is to make global, three-dimensional measurements of ozone and other chemical species involved in its formation, destruction, and distribution.

The primary location for obtaining TES data is NASA's Reverb Tool (http://reverb.echo.nasa.gov/). The TES Search and Subsetting Application (https://subset.larc.nasa.gov/tes) allows users to search for and subset TES datasets stored at the NASA Langley Atmospheric Science Data Center (ASDC). The web interface allows for selection criteria for desired species by date range(s) and/or geospatial coverage.

Appendix B

Administrative divisions included in analysis:

CN-11 Beijing 北京 Municipality

CN-12 Tianjin 天津 Municipality

CN-13 Hebei 河北 Province

CN-14 Shanxi 山西 Province

CN-15 Nei Mongol (mn) (Inner Mongolia) 内蒙古 Autonomous region

CN-21 Liaoning 辽宁 Province

CN-22 Jilin 吉林 Province

CN-23 Heilongjiang 黑龙江 Province

CN-31 Shanghai 上海 Municipality

CN-32 Jiangsu 江苏 Province

CN-33 Zhejiang 浙江 Province

CN-34 Anhui 安徽 Province

CN-35 Fujian 福建 Province

CN-36 Jiangxi 江西 Province

CN-37 Shandong 山东 Province

CN-41 Henan 河南 Province

CN-42 Hubei 湖北 Province

CN-43 Hunan 湖南 Province

CN-44 Guangdong 广东 Province

CN-45 Guangxi 广西 Autonomous region

CN-46 Hainan 海南 Province

CN-50 Chongqing 重庆 Municipality

CN-51 Sichuan 四川 Province

CN-52 Guizhou 贵州 Province

CN-53 Yunnan 云南 Province

CN-54 Xizang (Tibet) 西藏 Autonomous region

CN-61 Shaanxi 陕西 Province

CN-62 Gansu 甘肃 Province

CN-63 Qinghai 青海 Province

CN-64 Ningxia 宁夏 Autonomous region

CN-65 Xinjiang 新疆 Autonomous region